Causal Inference and ML: Beyond Correlation in Energy Data

Heterogeneous Causal Effects of Efficient Windows on U.S. Household Energy Use

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Project status | Module 07 Project Session

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Real-data project · EIA RECS 2020 · n = 17,196 households

Status presentation · full causal results coming in the final session

Analytical Framework: Efficient Windows -> Energy Use

Data & Variables

Treatment (T):

Efficient_Windows (1/0)

(double/triple vs single-pane)

Outcome (Y):

Energy_Intensity kWh/m2.a

Features (X) / Confounders:

Building age, Floor area,

Climate zone (8 IECC),

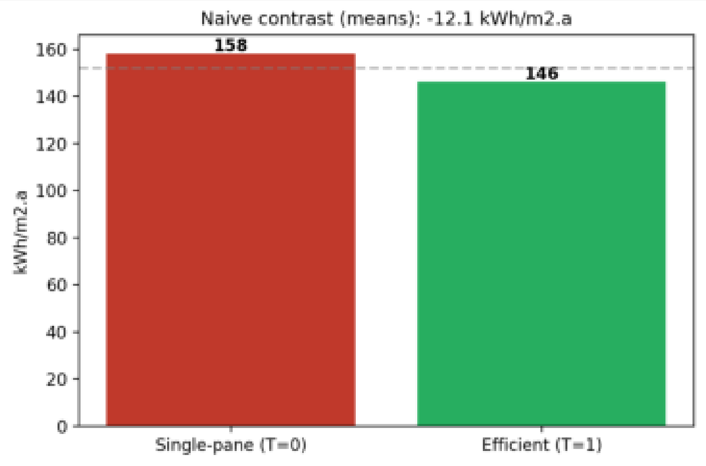
Housing type, HH size, Income

Data: EIA RECS 2020

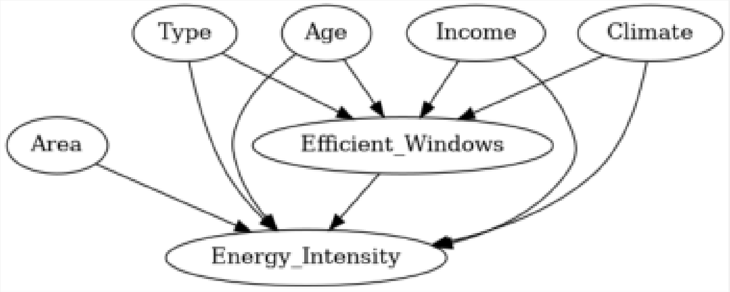
n = 17,196 U.S. households

Public domain (eia.gov)

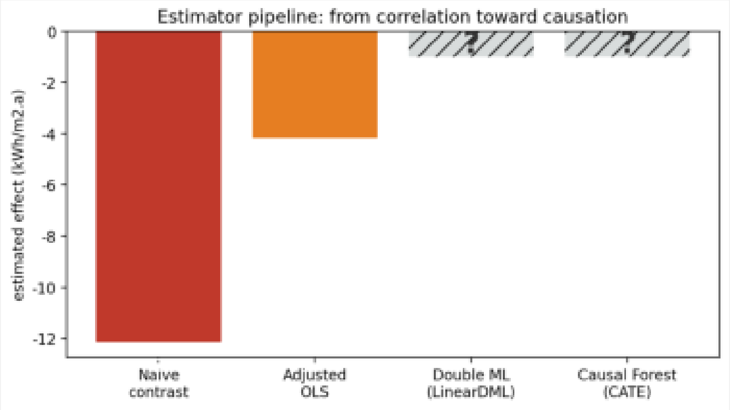
Naive Contrast (means)



Structural Model (DAG)



Estimator Pipeline (in progress)



Status & Next Steps

Naive: -12 · Adj. OLS: -4.2

Big shrink already from confounders.

Next - Double ML:

residualise Y and T with random forests, cross-fit -> remove ALL observable confounding.

Then - Causal Forest:

per-household effect tau(X) -> show which buildings benefit most.

Final ATE + CATE + targeting recommendation in next session.